Question 1

Planning (15 marks)

Defining the problem (3 marks)

P1 *d* is the independent variable or vary *d* (allow in table if numbers given) [1]

P2 *R* is the dependent variable or measure *R* as *d* varied (allow in table) [1]

P3 Keep output of light source constant (allow constant current / e.m.f. / voltage / power) [1]

Methods of data collection (5 marks)

M1 Diagram showing an LDR in a circuit and an independent lamp. [1]

M2 Diagram showing ruler measuring appropriate distance or *d* labelled correctly. [1]

M3 Correct circuit diagram for LDR using conventional symbols; allow labelled diagram [1]
Ammeter and voltmeter with power supply, or potential divider methods
ohmmeter without power supply, or
bridge methods.

M4 Method of determining *R*. [1]
Ohmmeter.
$R = V/I$ justified.
Potential divider equation
Description of balancing bridge with correct equation.

M5 Perform experiment in a dark room/tube [1]

Method of analysis (2 marks)

A1 Plot a graph of log *R* against log *d* [1]

A2 Relationship is correct if log *R* against log *d* graph is a straight line [1]

Safety considerations (1 mark)

S1 Do not look directly at bright light source / do not touch hot light source. [1]
Allow safety glasses with reference to light source.

Additional detail (4 marks)

D1/2/3/4 Relevant points might include [4]
Detail on measuring the distance
Keep orientation of LDR with respect to the light source constant
Reasoned method for keeping light and LDR in correct orientation. (E.g. use of set square, fix to rule, optical bench or equivalent)
Determination of a typical current
Range of ammeter / ohmmeter
Control (or monitoring) of an additional variable e.g. temperature
Reason for performing experiment in a dark room related to the LDR
Method for checking the output of the light source is constant.
Identifies gradient = *n* and/or *y*-intercept = log *k* for log *R* against log *d* graph

Do not allow parallax when reading ruler, or reflectors.

[Total: 15]

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